

SET 11 – Product Category Analysis

Q1. Pie Chart

```
category <- c("Electronics","Clothing","Appliances")
```

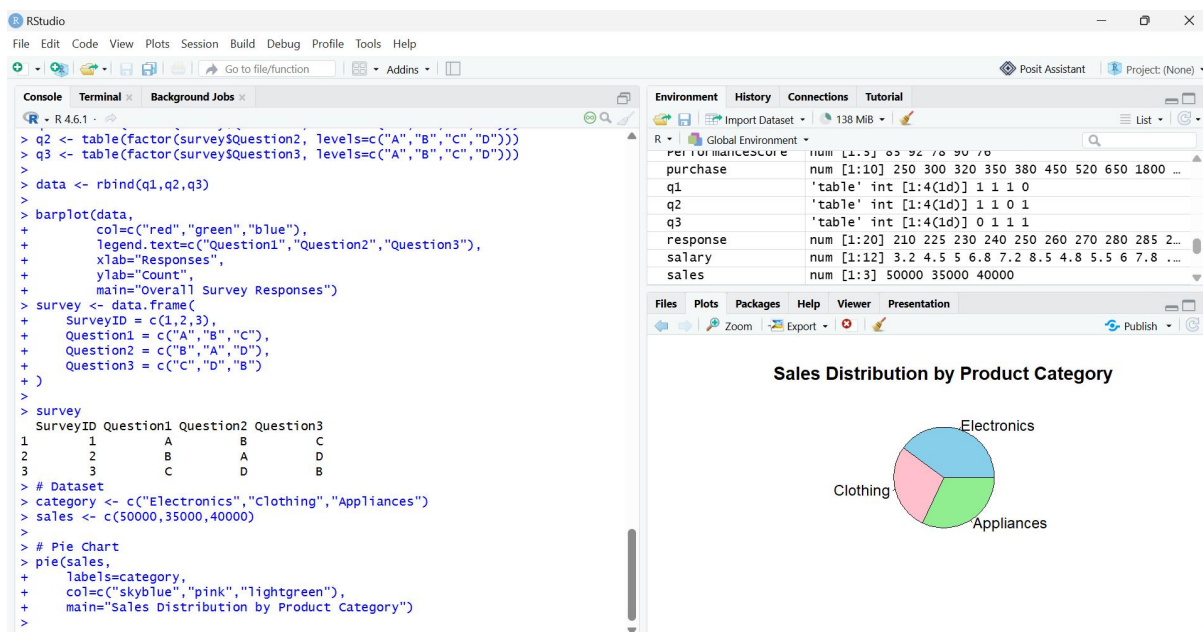
```
sales <- c(50000,35000,40000)
```

```
pie(sales,
```

```
  labels=category,
```

```
  col=c("skyblue","pink","lightgreen"),
```

```
  main="Sales Distribution by Product Category")
```



Q2. Funnel Chart

```
category <- c("Electronics","Clothing","Appliances")
```

```
sales <- c(50000,35000,40000)
```

```
par(mar=c(5,8,4,2))
```

```
barplot(rev(sales),
```

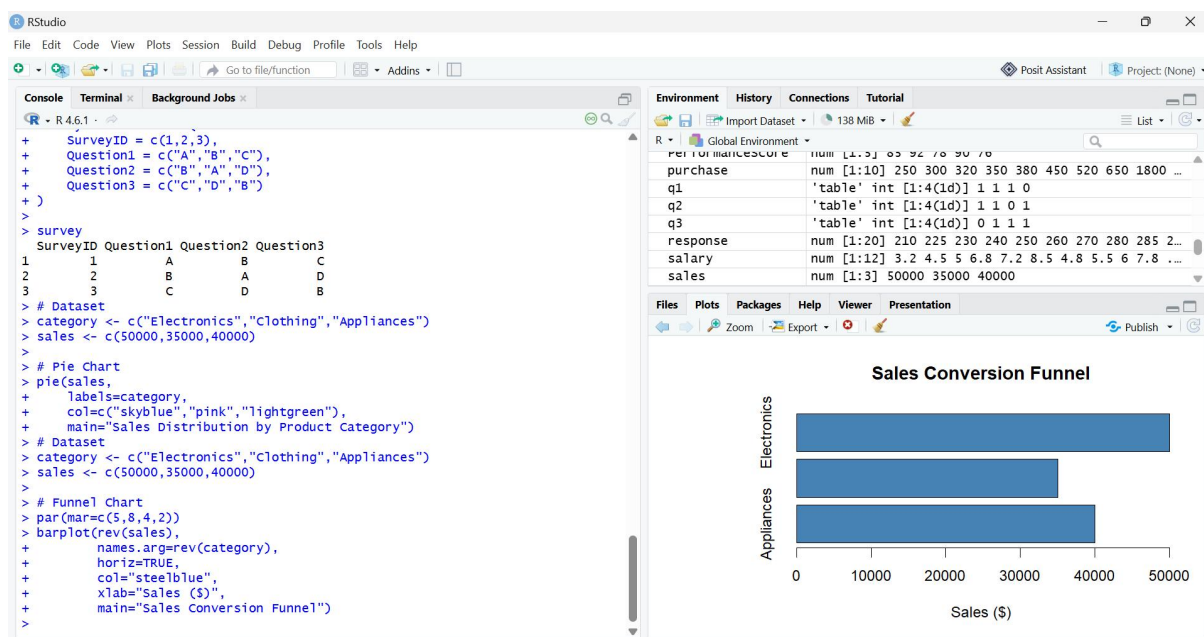
```
  names.arg=rev(category),
```

```
  horiz=TRUE,
```

```
  col="steelblue",
```

```
  xlab="Sales ($)"),
```

```
main="Sales Conversion Funnel")
```



Q3. Table

```
product <- data.frame(
  Category=c("Electronics","Clothing","Appliances"),
  Sales=c(50000,35000,40000)
)
```

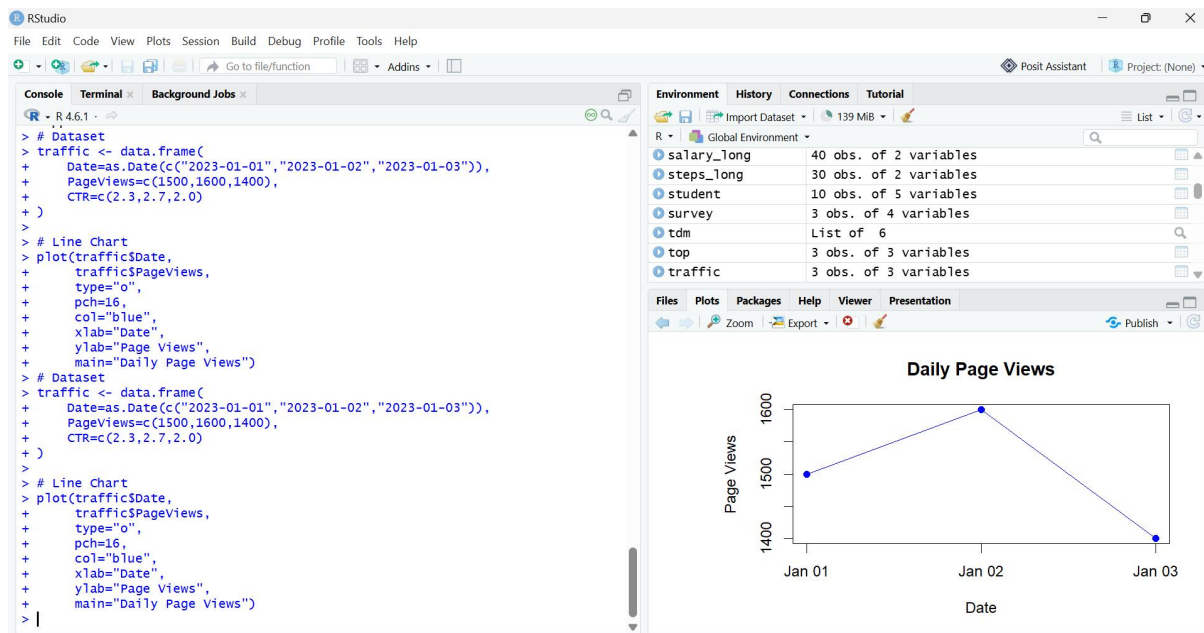
Product

```
R
> # Dataset
> product <- data.frame(
+   Category=c("Electronics","Clothing","Appliances"),
+   Sales=c(50000,35000,40000)
+ )
>
> product
  Category Sales
1 Electronics 50000
2  Clothing 35000
3 Appliances 40000
> |
```

SET 12 – Website Traffic Analysis

Q1. Line Chart

```
traffic <- data.frame(  
  Date=as.Date(c("2023-01-01","2023-01-02","2023-01-03")),  
  PageViews=c(1500,1600,1400),  
  CTR=c(2.3,2.7,2.0)  
)  
  
plot(traffic$Date,  
     traffic$PageViews,  
     type="o",  
     pch=16,  
     col="blue",  
     xlab="Date",  
     ylab="Page Views",  
     main="Daily Page Views")
```



Q2. Bar Chart

```
traffic <- data.frame(  
  Date=c("2023-01-01","2023-01-02","2023-01-03"),
```

```

PageViews=c(1500,1600,1400),

CTR=c(2.3,2.7,2.0)

)

barplot(traffic$CTR,

      names.arg=traffic$Date,

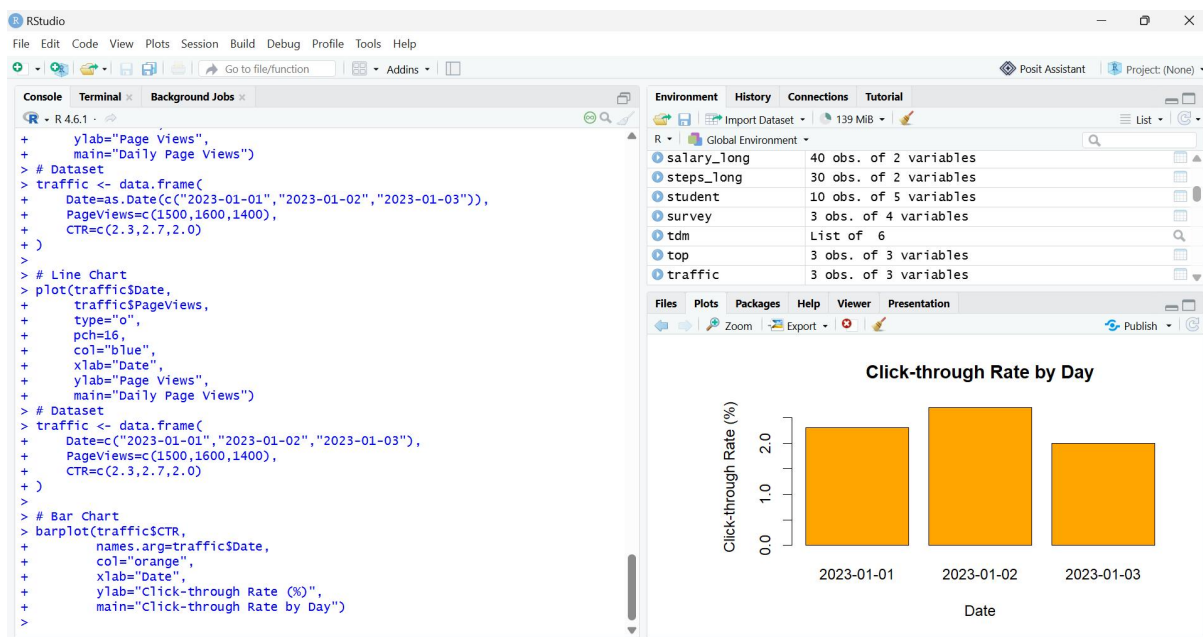
      col="orange",

      xlab="Date",

      ylab="Click-through Rate (%)",

      main="Click-through Rate by Day")

```



Q3. Table

```

traffic <- data.frame(

  Date=c("2023-01-01","2023-01-02","2023-01-03"),

  PageViews=c(1500,1600,1400),

  CTR=c("2.3%","2.7%","2.0%")

)

Traffic

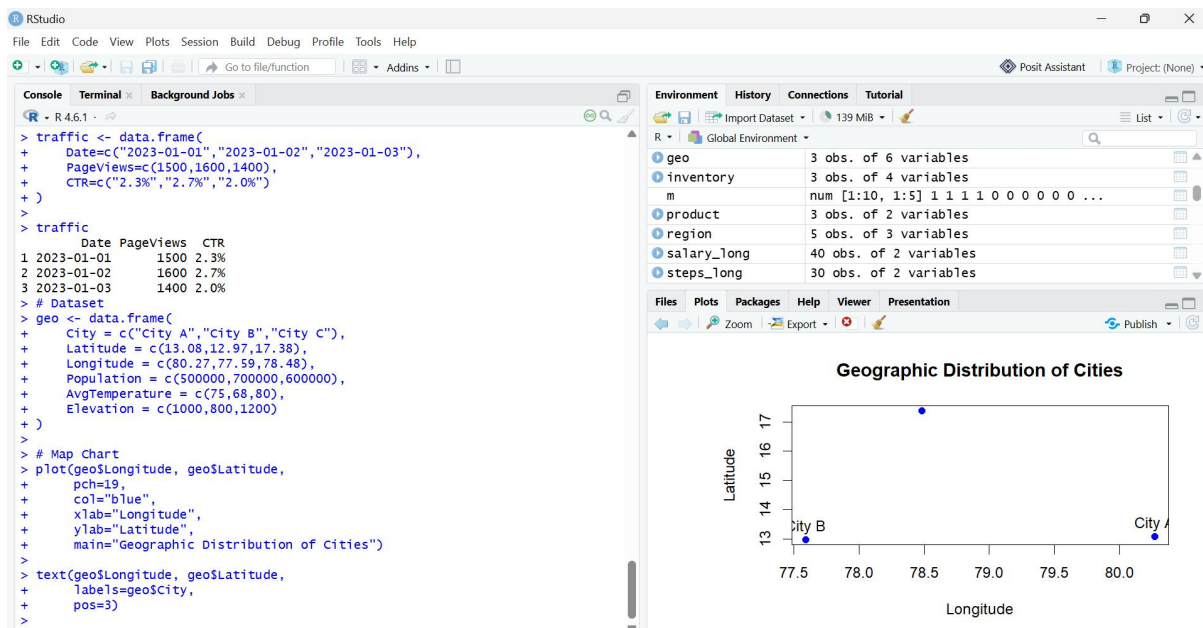
```

```
.  
> # Dataset  
> traffic <- data.frame(  
+   Date=c("2023-01-01","2023-01-02","2023-01-03"),  
+   PageViews=c(1500,1600,1400),  
+   CTR=c("2.3%","2.7%","2.0%")  
+ )  
>  
> traffic  
      Date PageViews  CTR  
1 2023-01-01      1500 2.3%  
2 2023-01-02      1600 2.7%  
3 2023-01-03      1400 2.0%  
> |
```

SET 13 – Geographic Data Analysis

Q1. Map Chart

```
geo <- data.frame(  
  City = c("City A","City B","City C"),  
  Latitude = c(13.08,12.97,17.38),  
  Longitude = c(80.27,77.59,78.48),  
  Population = c(500000,700000,600000),  
  AvgTemperature = c(75,68,80),  
  Elevation = c(1000,800,1200)  
)  
plot(geo$Longitude, geo$Latitude,  
  pch=19,  
  col="blue",  
  xlab="Longitude",  
  ylab="Latitude",  
  main="Geographic Distribution of Cities")  
text(geo$Longitude, geo$Latitude,  
  labels=geo$City,  
  pos=3)
```

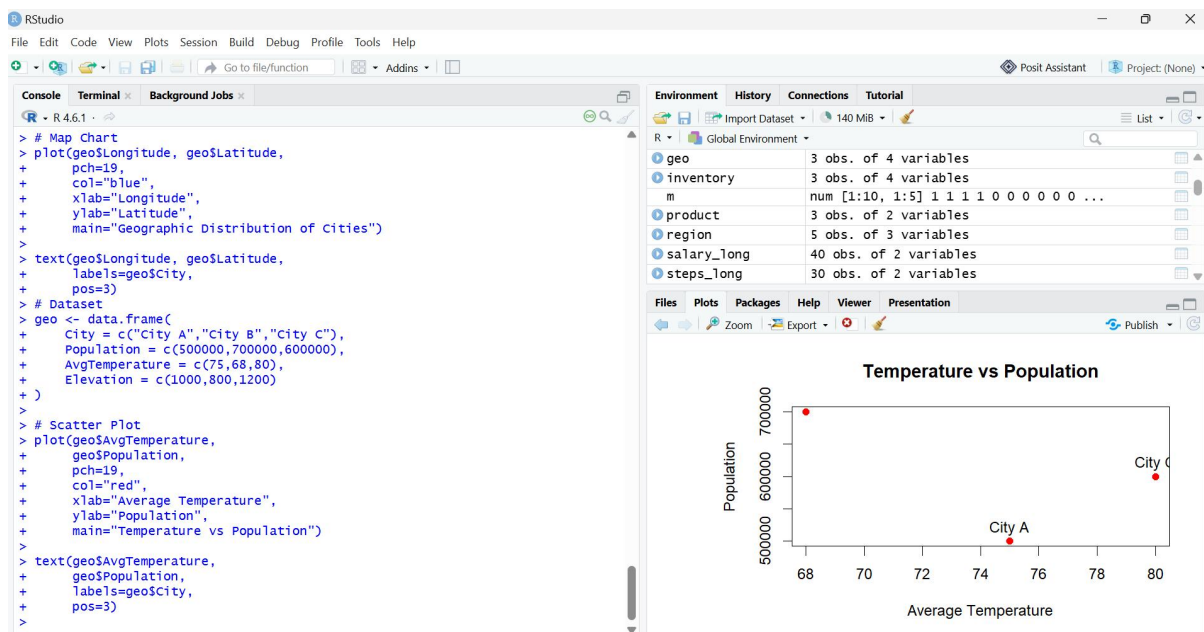


Q2. Scatter Plot

```
geo <- data.frame(  
  City = c("City A","City B","City C"),  
  Population = c(500000,700000,600000),  
  AvgTemperature = c(75,68,80),  
  Elevation = c(1000,800,1200)  
)
```

```
plot(geo$AvgTemperature,  
     geo$Population,  
     pch=19,  
     col="red",  
     xlab="Average Temperature",  
     ylab="Population",  
     main="Temperature vs Population")
```

```
text(geo$AvgTemperature,  
     geo$Population,  
     labels=geo$City,  
     pos=3)
```



Q3. Table

```
geo <- data.frame(  
  City = c("City A","City B","City C"),  
  Population = c(500000,700000,600000),  
  AvgTemperature = c(75,68,80),  
  Elevation = c(1000,800,1200)  
)
```

Geo

```
> # Dataset  
> geo <- data.frame(  
+   City = c("City A","City B","City C"),  
+   Population = c(500000,700000,600000),  
+   AvgTemperature = c(75,68,80),  
+   Elevation = c(1000,800,1200)  
+ )  
>  
> geo  
  City Population AvgTemperature Elevation  
1 City A      5e+05           75      1000  
2 City B      7e+05           68       800  
3 City C      6e+05           80      1200  
>
```


SET 14 – Survey Results Analysis

Q1. Stacked Bar Chart

```
survey <- data.frame(

  Respondent = c(1,2,3),

  Question1 = c("A","B","C"),

  Question2 = c("B","A","D"),

  Question3 = c("C","D","B")

)

q1 <- table(factor(survey$Question1, levels=c("A","B","C","D")))

barplot(rbind(q1),

  beside=FALSE,

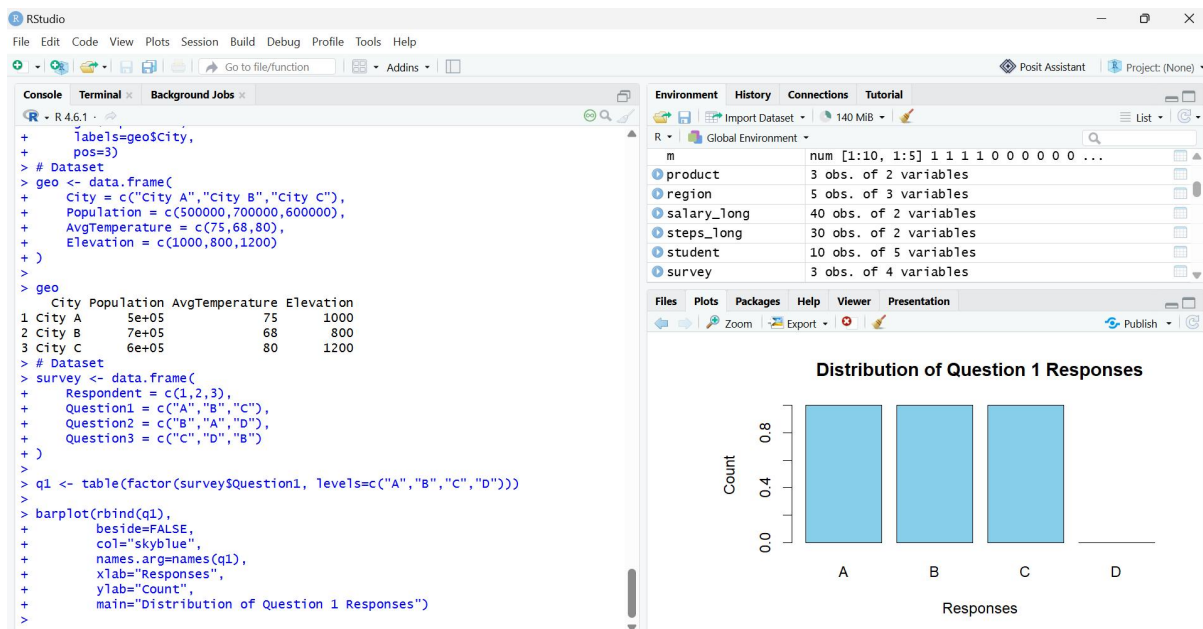
  col="skyblue",

  names.arg=names(q1),

  xlab="Responses",

  ylab="Count",

  main="Distribution of Question 1 Responses")
```



Q2. Radar Chart

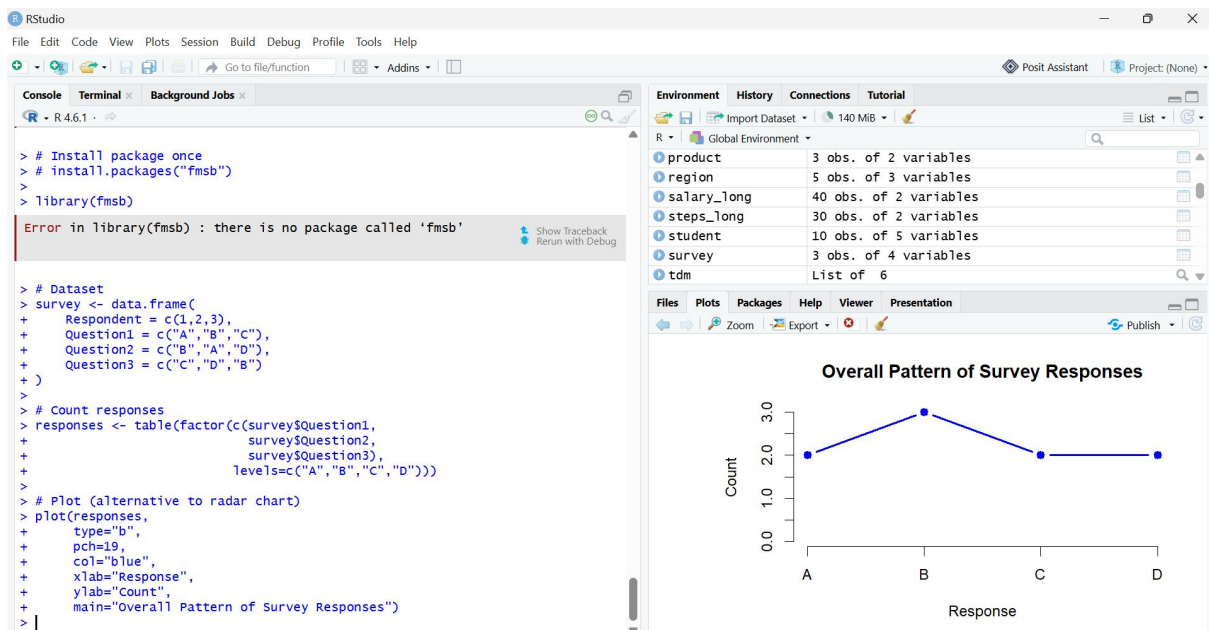
```
survey <- data.frame(

  Respondent = c(1,2,3),
```

```

Question1 = c("A","B","C"),
Question2 = c("B","A","D"),
Question3 = c("C","D","B")
)
responses <- table(factor(c(survey$Question1,
                           survey$Question2,
                           survey$Question3),
                           levels=c("A","B","C","D"))))
plot(responses,
     type="b",
     pch=19,
     col="blue",
     xlab="Response",
     ylab="Count",
     main="Overall Pattern of Survey Responses")

```



Q3. Table

```

survey <- data.frame(
  Respondent = c(1,2,3),
  Question1 = c("A","B","C"),

```

```
Question2 = c("B","A","D"),
```

```
Question3 = c("C","D","B")
```

```
)
```

Survey

```
> # Dataset
> survey <- data.frame(
+   Respondent = c(1,2,3),
+   Question1 = c("A","B","C"),
+   Question2 = c("B","A","D"),
+   Question3 = c("C","D","B")
+ )
>
> survey
  Respondent Question1 Question2 Question3
1          1         A         B         C
2          2         B         A         D
3          3         C         D         B
> |
```

SET 15 – Survey Responses Analysis

Q17. Grouped Bar Chart

```
survey <- data.frame(
```

```
  SurveyID = c(1,2,3),
```

```
  Question1 = c("A","B","C"),
```

```
  Question2 = c("B","A","D"),
```

```
  Question3 = c("C","D","B")
```

```
)
```

```
q1 <- table(survey$Question1)
```

```
barplot(q1,
```

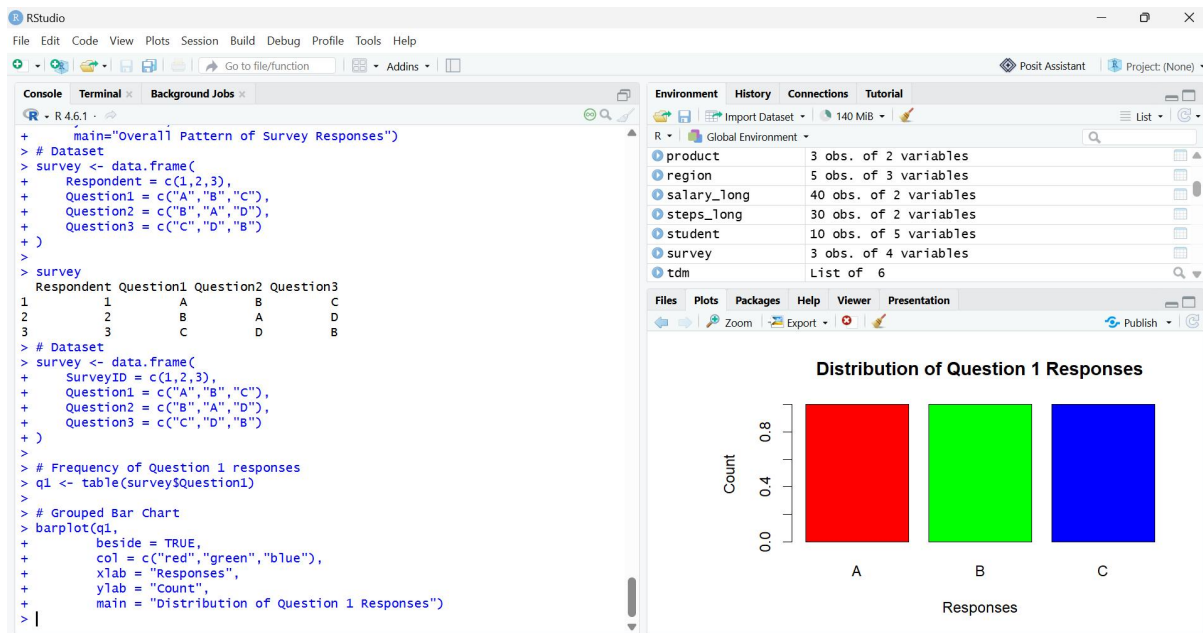
```
  beside = TRUE,
```

```
  col = c("red","green","blue"),
```

```
  xlab = "Responses",
```

```
  ylab = "Count",
```

```
  main = "Distribution of Question 1 Responses")
```



Q18. Stacked Bar Chart

```

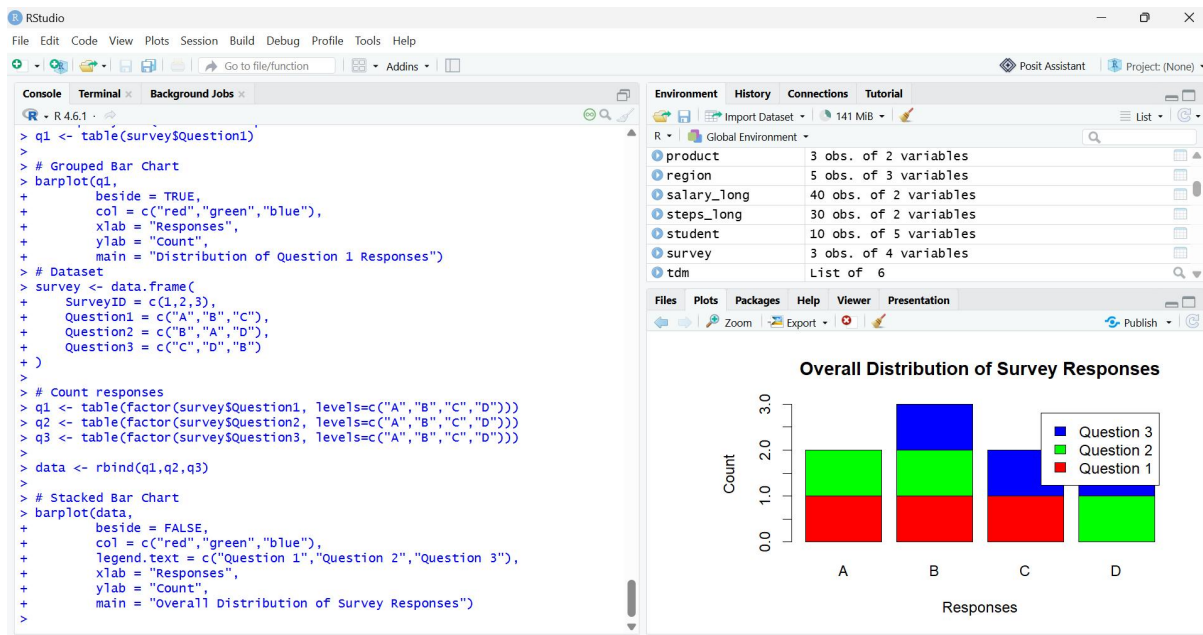
survey <- data.frame(
  SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","D"),
  Question3 = c("C","D","B")
)

q1 <- table(factor(survey$Question1, levels=c("A","B","C","D")))
q2 <- table(factor(survey$Question2, levels=c("A","B","C","D")))
q3 <- table(factor(survey$Question3, levels=c("A","B","C","D")))

data <- rbind(q1,q2,q3)

barplot(data,
  beside = FALSE,
  col = c("red","green","blue"),
  legend.text = c("Question 1","Question 2","Question 3"),
  xlab = "Responses",
  ylab = "Count",
  main = "Overall Distribution of Survey Responses")

```



Q19. Table

```
survey <- data.frame(
```

```
  SurveyID = c(1,2,3),
```

```
  Question1 = c("A","B","C"),
```

```
  Question2 = c("B","A","D"),
```

```
  Question3 = c("C","D","B")
```

```
)
```

```
Survey
```

```

> # Dataset
> survey <- data.frame(
+   SurveyID = c(1,2,3),
+   Question1 = c("A","B","C"),
+   Question2 = c("B","A","D"),
+   Question3 = c("C","D","B")
+ )
> # Table
> survey
  SurveyID Question1 Question2 Question3
1         1         A         B         C
2         2         B         A         D
3         3         C         D         B
>

```